

Exercise7

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```
library(dplyr) # A Grammar of Data Manipulation
library(ggplot2) # Create Elegant Data Visualisations Using the Grammar of Graphics
library(kableExtra) # Construct Complex Table with 'kable' and Pipe Syntax
library(mlogit) # Multinomial Logit Models
```

```
data("Heating",
      package = "mlogit")
```

```
H <- mlogit.data(Heating,
                 shape = "wide",
                 choice = "depvar",
                 varying = c(3:12))
```

1. Restate the blue bus-red bus situation as a nested logit model. What are the marginal and conditional probabilities of this model?

The nested model organizes choices hierarchically. At the top level, individuals choose between two nests - either Car or Bus Nest. The Bus Nest contains the correlated alternatives: Blue Bus (bb) and Red Bus (rb). The dissimilarity parameter (λ) quantifies the degree of correlation within the bus nest. If λ tends to 1, it would indicate the correlation between the variables within the nests is zero.

First, define the aggregates utilities of the bus alternatives: $e^{IV_{bus}}$

Then the probabilities of choosing car:

$$P_{car} = \frac{e^{V_{car}}}{e^{V_{car}} + e^{IV_{bus}}}$$

The probabilities of choosing car:

$$P_{bus} = \frac{e^{IV_{bus}}}{e^{V_{car}} + e^{IV_{bus}}}$$

Conditional Probabilities

Probability of Blue Bus in the Bus Nest:

$$P_{bb|bus} = \frac{e^{V_{bb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Probability of Red Bus in the Bus Nest:

$$P_{rb|bus} = \frac{e^{V_{rb}/\lambda}}{e^{V_{bb}/\lambda} + e^{V_{rb}/\lambda}}$$

Marginal Probabilities

For the blue bus:

$$P_{bb} = P_{bus} * P_{bb|bus}$$

For the red bus:

$$P_{rb} = P_{bus} * P_{rb|bus}$$

2. Use model nl2 in this chapter and calculate the direct-point elasticity at the mean

values of the variables, for an increase in the installation costs of Gas Central systems.

```
nl2 <- mlogit(depvar ~ ic + oc, H,
  nests = list(room = c('er', 'gr'), central = c('ec', 'gc', 'hp')),
  # this will set all lambdas to be the same
  un.nest.el = TRUE,
  steptol = 1e-12)
summary(nl2)
```

```
##
## Call:
## mlogit(formula = depvar ~ ic + oc, data = H, nests = list(room = c("er",
##      "gr"), central = c("ec", "gc", "hp")), un.nest.el = TRUE,
##      steptol = 1e-12)
##
## Frequencies of alternatives:choice
##      ec      er      gc      gr      hp
## 0.071111 0.093333 0.636667 0.143333 0.055556
##
## bfgs method
## 19 iterations, 0h:0m:0s
## g'(-H)^-1g = 0.0443
## successive function values within tolerance limits
##
## Coefficients :
##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept):er -1.1290e+00  7.8672e-02 -14.3508 < 2.2e-16 ***
## (Intercept):gc -1.0474e-02  1.5358e-02  -0.6820 0.4952521
## (Intercept):gr -1.1817e+00  8.0048e-02 -14.7628 < 2.2e-16 ***
## (Intercept):hp -4.5459e-02  1.5516e-02  -2.9298 0.0033921 **
## ic              -8.3899e-05  2.5826e-05  -3.2486 0.0011598 **
## oc              -2.2841e-04  5.9509e-05  -3.8383 0.0001239 ***
## iv              2.7374e-02  3.2338e-03   8.4649 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -1003.5
## McFadden R^2:  0.018329
## Likelihood ratio test : chisq = 37.473 (p.value = 3.6549e-08)
```

```
X_mean <- model.matrix(nl2)[1:5,]
alt <- index(H)$alt[1:5]
```

```
mean_ic <- H %>%
  group_by(alt) %>%
```

```

summarize(ic = mean(ic)) %>%
arrange(alt)

mean_oc <- H %>%
  group_by(alt) %>%
  summarize(oc = mean(oc)) %>%
  arrange(alt)

X_mean[,5] <- mean_ic$ic
X_mean[,6] <- mean_oc$oc

# Electric central
exp_V_ec <-
  exp((X_mean[alt == c("ec"), "oc"] * coef(nl2)["oc"] +
        X_mean[alt == c("ec"), "ic"] * coef(nl2)["ic"])
        / coef(nl2)["iv"])
# Electric room
exp_V_er <-
  exp((coef(nl2)["(Intercept):er"] +
        X_mean[alt == c("er"), "oc"] * coef(nl2)["oc"] +
        X_mean[alt == c("er"), "ic"] * coef(nl2)["ic"])
        / coef(nl2)["iv"])
# Gas central
exp_V_gc <-
  exp((coef(nl2)["(Intercept):gc"] +
        X_mean[alt == c("gc"), "oc"] * coef(nl2)["oc"] +
        X_mean[alt == c("gc"), "ic"] * coef(nl2)["ic"])
        / coef(nl2)["iv"])
# Gas room
exp_V_gr <-
  exp((coef(nl2)["(Intercept):gr"] +
        X_mean[alt == c("gr"), "oc"] * coef(nl2)["oc"] +
        X_mean[alt == c("gr"), "ic"] * coef(nl2)["ic"])
        / coef(nl2)["iv"])
# Heat pump
exp_V_hp <-
  exp((coef(nl2)["(Intercept):hp"] +
        X_mean[alt == c("hp"), "oc"] * coef(nl2)["oc"] +
        X_mean[alt == c("hp"), "ic"] * coef(nl2)["ic"])
        / coef(nl2)["iv"])

# Conditional probabilities of systems within the central nest
cp_c <- data.frame(ec = exp_V_ec / (exp_V_ec + exp_V_gc + exp_V_hp),
                  gc = exp_V_gc / (exp_V_ec + exp_V_gc + exp_V_hp),
                  hp = exp_V_hp / (exp_V_ec + exp_V_gc + exp_V_hp))

# Conditional probabilities of systems within the room nest
cp_r <- data.frame(er = exp_V_er / (exp_V_er + exp_V_gr),
                  gr = exp_V_gr / (exp_V_er + exp_V_gr))

#After removing ec
mp <- data.frame(central = exp(coef(nl2)["iv"] * log(exp_V_ec + exp_V_gc + exp_V_hp))
                  / (exp(coef(nl2)["iv"] * log(exp_V_ec + exp_V_gc + exp_V_hp)) +

```

```

        exp((coef(nl2)["iv"] * log(exp_V_er + exp_V_gr)))),
room = exp(coef(nl2)["iv"] * log(exp_V_er + exp_V_gr)) /
(exp(coef(nl2)["iv"] * log(exp_V_gc + exp_V_hp)) +
exp((coef(nl2)["iv"] * log(exp_V_er + exp_V_gr))))))

nlp <- data.frame(system = c("ec", "er", "gc", "gr", "hp"),
  # Conditional probability
  cp = c(cp_c$ec, cp_r$er, cp_c$gc, cp_r$gr, cp_c$hp),
  # Marginal probability
  mp = c(mp$central, mp$room, mp$central, mp$room, mp$central),
  beta_ic = c(as.numeric(nl2$coefficients["ic"])),
  beta_oc = c(as.numeric(nl2$coefficients["oc"])),
  lambda = c(1 - as.numeric(nl2$coefficients["iv"]))) %>%
  # Joint probability
  mutate(p = cp * mp)

nlp <- cbind(nlp, X_mean[,5:6]) %>%
  # Increase installation cost 1%
  mutate(ic_1pct = if_else(nlp$system == "gc", 1.01 * ic, ic))

direct_elasticities <- nlp %>%
  transmute(DEM = ((1 - mp) + (1 - cp) * (1 - lambda)/lambda) * beta_ic * ic)

direct_elasticities

##          DEM
## 1 -0.01814786
## 2 -0.06446512
## 3 -0.01570029
## 4 -0.05979788
## 5 -0.02306906

```

3. Use model n12 in this chapter and calculate the cross-point elasticity at the mean values of the variables, for a 1% increase in the operation costs of Gas Central

systems.

```

elasticities <- nlp %>%
  transmute(CEM_ec = -(mp + (1 - lambda)/lambda * cp) * beta_ic * mean_ic$ic[mean_ic == "ec"]
  CEM_er = -(mp + (1 - lambda)/lambda * cp) * beta_ic * mean_ic$ic[mean_ic == "er"]
  CEM_gc = -(mp + (1 - lambda)/lambda * cp) * beta_ic * mean_ic$ic[mean_ic == "gc"]
  CEM_gr = -(mp + (1 - lambda)/lambda * cp) * beta_ic * mean_ic$ic[mean_ic == "gr"]
  CEM_hp = -(mp + (1 - lambda)/lambda * cp) * beta_ic * mean_ic$ic[mean_ic == "hp"]
  # Transmute so that each row is the elasticity due to changes to a system
  t()

diag(elasticities) <- direct_elasticities$DEM

elasticities

##          1          2          3          4          5
## CEM_ec -0.01814786  0.01710267  0.05446052  0.01763471  0.05294864
## CEM_er  0.06321789 -0.06446512  0.06498776  0.02104351  0.06318363
## CEM_gc  0.04991152  0.01611293 -0.01570029  0.01661418  0.04988447
## CEM_gr  0.05922422  0.01911935  0.06088228 -0.05979788  0.05919213

```

```
## CEM_hp 0.06723697 0.02170610 0.06911936 0.02238135 -0.02306906
```

4. Re-estimate the nested logit model in this chapter, but change the nests to types of energy as follows: - Gas: gas central, gas room. - Electricity: electric central, electric room, heat pump. Use a single coefficient for the inclusive variables (i.e., set `un.nest.el = TRUE`). Are the results reasonable? Discuss.

```
nl1 <- mlogit(depvar ~ oc + ic, H,
              nests = list(gas = c('gc', 'gr'),
                           electrical = c('ec', 'er', 'hp')),
              un.nest.el = TRUE,
              steptol = 1e-12)
summary(nl1)
```

```
##
## Call:
## mlogit(formula = depvar ~ oc + ic, data = H, nests = list(gas = c("gc",
## "gr"), electrical = c("ec", "er", "hp")), un.nest.el = TRUE,
## steptol = 1e-12)
##
## Frequencies of alternatives:choice
##      ec      er      gc      gr      hp
## 0.071111 0.093333 0.636667 0.143333 0.055556
##
## bfgs method
## 14 iterations, 0h:0m:0s
## g'(-H)^-1g = 5.36E-07
## gradient close to zero
##
## Coefficients :
##              Estimate Std. Error z-value Pr(>|z|)
## (Intercept):er  0.8391327  0.6751569  1.2429 0.213916
## (Intercept):gc  1.0431426  1.4889576  0.7006 0.483562
## (Intercept):gr -3.1044881  1.3014025 -2.3855 0.017056 *
## (Intercept):hp -2.4162407  0.8791472 -2.7484 0.005989 **
## oc              -0.0095729  0.0023864 -4.0114 6.035e-05 ***
## ic              -0.0030007  0.0010566 -2.8398 0.004514 **
## iv              2.9493296  1.5797005  1.8670 0.061899 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Log-Likelihood: -1005.3
## McFadden R^2: 0.016517
## Likelihood ratio test : chisq = 33.768 (p.value = 2.218e-07)
```

The re-estimated nested logit model groups alternatives by energy type with a single inclusive value coefficient. Both cost variables (`oc`, `ic`) are negative and statistically significant, aligning with expectations—higher costs reduce utility. However, as the `iv` parameter (2.949) exceeds 1, violating the nested logit model's theoretical requirement for consistency. This suggests improper nesting or misspecification.

Alternative-specific constants reveal lower baseline utility for gas room (`gr`) and heat pump (`hp`) compared to electric central (`ec`), but gas central (`gc`) and electric room (`er`) constants are insignificant. The model's poor fit (McFadden $R^2 = 0.017$) further questions its validity.